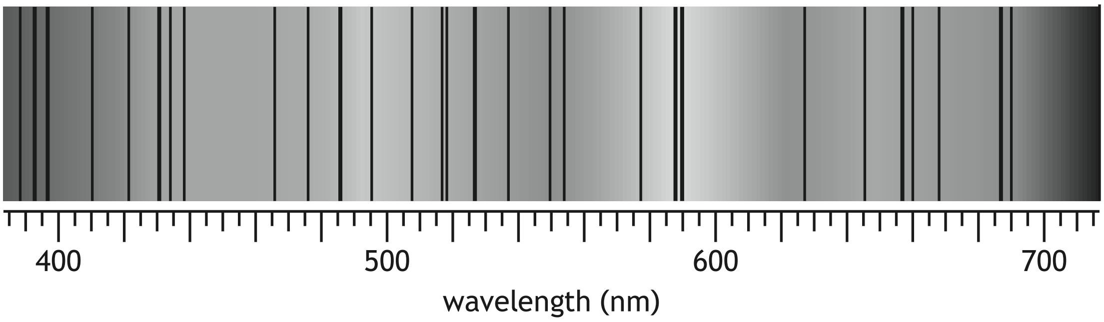
10. Light from the Sun is a continuous spectrum containing dark lines as shown.



(a) Explain how these dark lines in the spectrum of sunlight are produced. **2**

(b) One of the dark lines corresponds to the blue-green line in the hydrogen spectrum.

(i) State the wavelength of this blue-green spectral line. **1**

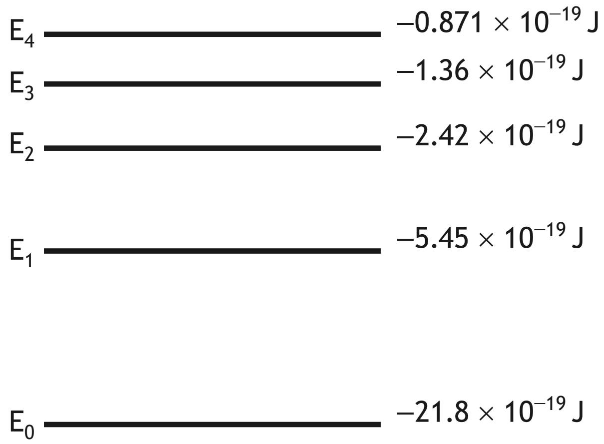
(ii) Calculate the frequency of this spectral line. **3**

Space for working and answer

10. (b) (continued)

MARKS | DO NOT WRITE IN THIS MARGIN

(iii) Some of the energy levels of the hydrogen atom are shown.



The dark line corresponding to the blue-green line in the hydrogen spectrum is produced due to an electron transition between E_1 and one of the other energy levels.

(A) Calculate the energy of the photon involved in this transition. **3**

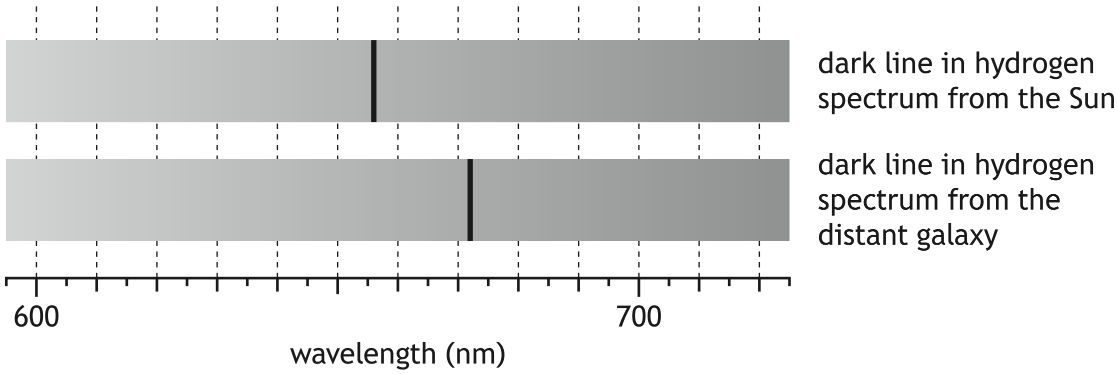
Space for working and answer

(B) Identify the electron transition that produces this dark line. **1**

[Turn over

10. (continued)

(c) One of the other dark lines in the spectrum of the Sun corresponds to a wavelength of 656 nm in the hydrogen spectrum. In the spectrum of light from a distant galaxy, the corresponding dark line is observed at 672 nm.



(i) Determine the recessional velocity of this galaxy. **5**

Space for working and answer

MARKS DO NOT WRITE IN THIS MARGIN

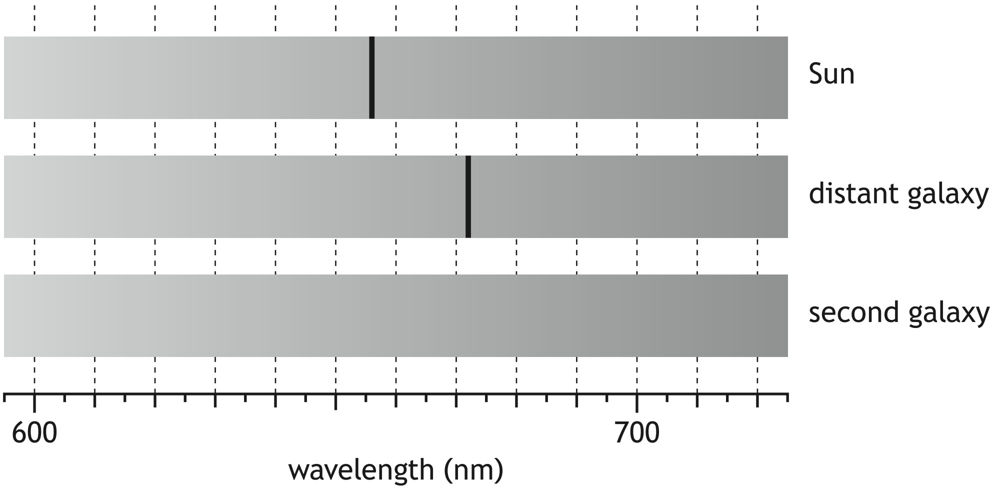
10. (c) (continued)

(ii) A second galaxy produces a spectrum where the corresponding dark line is observed in a different position. This galaxy has a recessional velocity of 4.88×10^6 m s^{-1}.

On the diagram below, add a line to show where this dark line would be observed on the spectrum of the second galaxy.

A numerical value is not required. **1**

(An additional diagram, if required, can be found on *page 52*.)



(d) Through observation of distant galaxies, it has been observed that the Universe is expanding at an accelerating rate.

State what physicists think is responsible for this accelerating rate of expansion. **1**

[Turn over